

local test

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1. Solve the equation $6x^{3/5} + 1 = 12/x^{3/5}$, giving your answers correct to 2 decimal places. [4 marks]

2. Write $(16 + 11\sqrt{10})/(2 + \sqrt{10}) + 1$ in the form $p + q\sqrt{10}$, where p and q are integers. Do not use a calculator. [4 marks]

3. Two similar triangles: smaller height $1+7\sqrt{5}$, larger height $a+b\sqrt{5}$. Base of smaller $4+2\sqrt{5}$, base of larger 20. Find a and b . [4 marks]

4. Solve the equation $12/x^{1/3} - x^{1/3} = 4$ [4 marks]
